

# HOSTEL GATEPASS MANAGEMENT SYSTEM

## Software Requirements Analysis Document

### 1. Problem Statement

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The Hostel Gatepass Management System is a proposed digital platform that automates and streamlines the process of granting, approving, tracking, and auditing student movement in and out of hostel premises. It replaces manual paper-based processes with a structured digital workflow, ensuring enhanced security, transparency, and accountability across all stakeholders.

### 2. Stakeholders & Use Case Diagram

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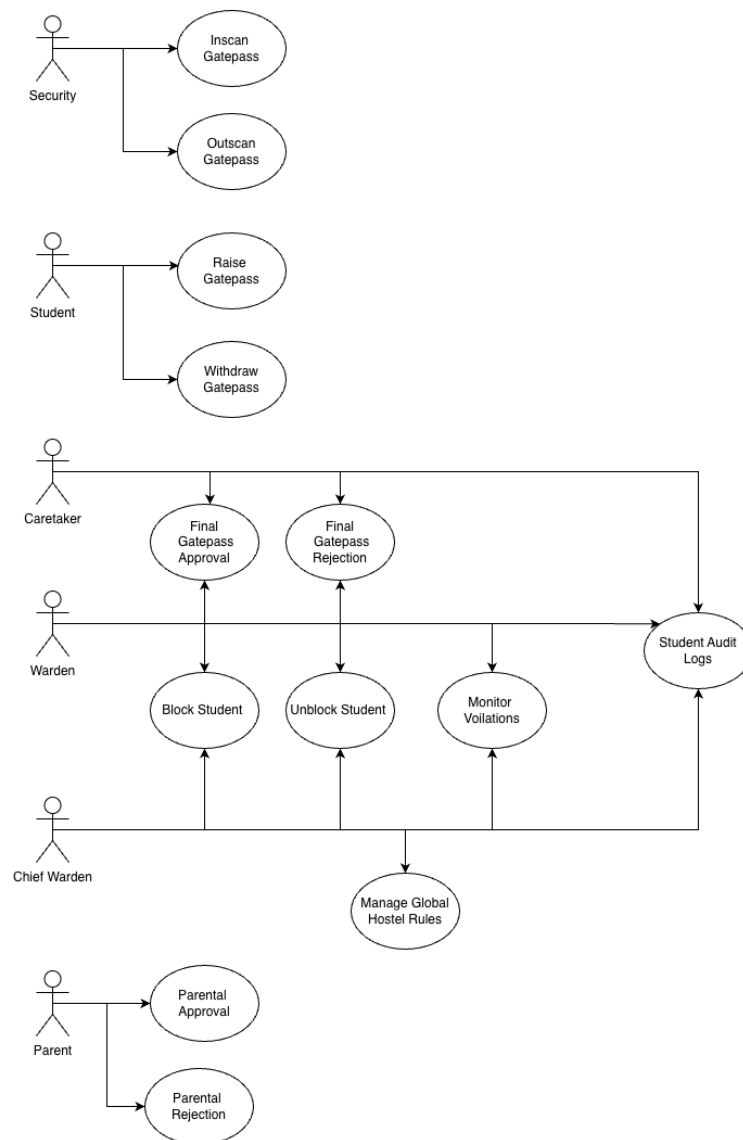
#### 2.1 Stakeholders

| Stakeholder  | Role                  | Key Responsibilities                                                          |
|--------------|-----------------------|-------------------------------------------------------------------------------|
| Student      | Primary User          | Raises, views, and withdraws gatepass requests. Presents QR code at the gate. |
| Parent       | External Approver     | Receives notification for Home Passes, approves or rejects via the portal.    |
| Caretaker    | Hostel-Block-Approver | Gives final approval or rejection for all pass types.                         |
| Warden       | Hostel Manager        | Blocks/unblocks students, monitors violations, views audit logs.              |
| Chief Warden | Institution Head      | Manages global hostel rules (curfew times, pass durations, advance notice).   |
| Security     | Gate Officer          | Scans QR code on student exit (Outscan) and return (Inscan).                  |

## 2.2 Use Case Diagram

The diagram below illustrates how each actor interacts with the system:

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## 3. Functional Requirements

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### 3.1 Role Management

- The system enforces strict role-based access control; each actor (Student, Parent, Caretaker, Warden, Chief Warden, Security) can only access screens and perform actions permitted to their role.
- No cross-role data visibility — a student cannot see another student's passes; a caretaker sees only their assigned block.

### 3.2 Gatepass Management

- Students can raise two types of passes: Day Pass and Home Pass.
- Day Pass: system auto-approves at caretaker level within seconds; QR code generated immediately; no parental approval needed.
- Home Pass: requires full approval chain — Student raises → Parent approves/rejects → Caretaker gives final approval; QR generated only after caretaker approval.
- Students can withdraw a pending pass before it is approved.
- Each pass records: destination, purpose, mode of transport, expected date/time, and actual return date/time.

### 3.3 QR Code & Gate Scanning

- A unique QR code is generated per approved pass.
- Security performs Outscan (exit) — pass status set to Active, exit timestamp recorded.
- Security performs Inscan (return) — pass status set to Completed, return timestamp recorded.
- QR code becomes invalid after Inscan or upon expiry of the approved return time.
- If a student returns late, a violation is automatically flagged in the system.

### 3.4 Pass Actions & Audit Trail

- Every state change on a pass — submission, approval, rejection, withdrawal, outscan, inscan — is recorded as a `Pass_Action` entry containing: actor ID, action type, timestamp, and optional remarks.
- Caretakers and Wardens can view the complete action history for any pass or any student under their block.

### 3.5 Violation & Blocking

- Every late return automatically increments the student's defaulter attempt count for the current semester.
- Once a student exhausts the configured maximum attempts (set by Chief Warden), the system automatically blocks them for the remainder of the semester — preventing any new pass from being raised.
- Wardens can manually block or unblock students for disciplinary reasons; Chief Warden holds override authority to unblock any student regardless of block type.
- Each block record stores: student, hostel block, blocking reason, blocking role, and blocked/unblocked timestamps.

### 3.6 Global Rule Configuration

The Chief Warden can set and update institution-wide rules at any time without code changes:

- Hostel opening time (earliest permitted exit)
- Hostel closing time / curfew (latest permitted return)
- Minimum advance notice required to raise a Home Pass (e.g., 2 days)
- Maximum allowed duration of a Home Pass (e.g., 7 days)

## 4. Non-Functional Requirements

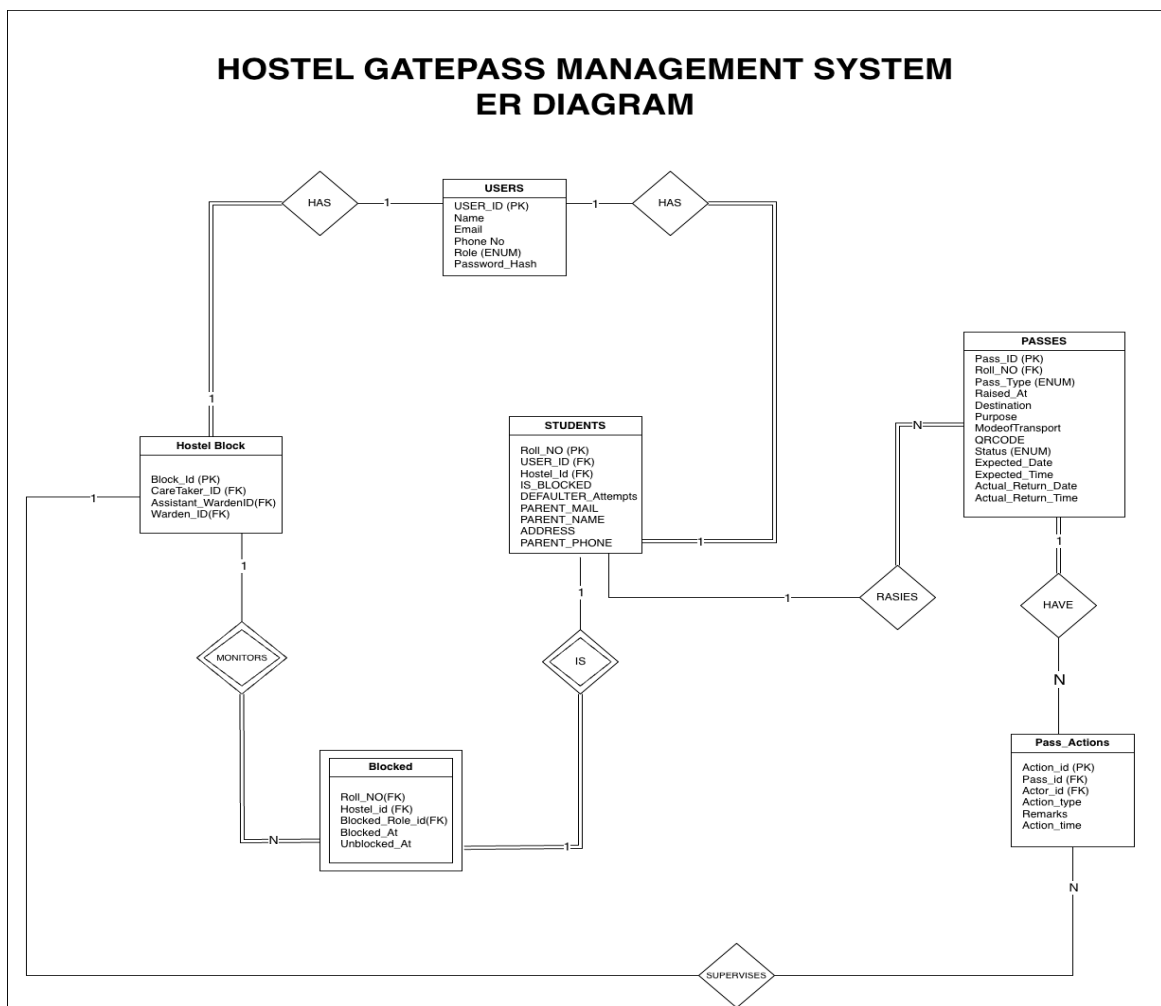
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| Category    | Requirement              | Detail                                                                                                                  |
|-------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Performance | Gate Scan Response       | Outscan and Inscan operations must reflect updated pass status within 2 seconds of scan.                                |
| Security    | QR Tamper Prevention     | Each QR code is cryptographically unique, single-use, and bound to a specific pass ID and student cant be reused.       |
| Reliability | Approval Chain Integrity | A pass must never skip an approval step system must enforce the correct sequence                                        |
| Scalability | Concurrent Users         | The system must handle all students of a hostel block submitting or scanning passes simultaneously without degradation. |

|                 |                    |                                                                                                                                                     |
|-----------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Usability       | Security Scan UX   | Security personnel must be able to complete a full gate scan in under 5 seconds with no training required.                                          |
| Maintainability | Configurable rules | Chief Warden must be able to update semester rules (defaulter limit, curfew, pass duration) through the UI without any code change or redeployment. |

## 5. Database Schema (ER Diagram)

### 5.1 ER Diagram



## 5.2 Entities Descriptions

| Entity       | Description                                                                                |
|--------------|--------------------------------------------------------------------------------------------|
| USERS        | Stores all system actors.                                                                  |
| STUDENTS     | Extends USERS. Each student is linked to a hostel block with their parent contact details. |
| Hostel Block | Each block has their respective authorities.                                               |
| PASSES       | Central table to have a Gate Pass description with status and pass type.                   |
| Pass_Actions | Audit log for every action on a pass. One pass can have N actions.                         |
| Blocked      | Records each block event for a student which role unblocked them and when.                 |

## 6. UI / Figma Designs

This section is for UI wireframes and high-fidelity designs produced in Figma.

<https://www.figma.com/design/qaRD994mBFCMEE7lZrisFc/Gatepass-UI?node-id=15-2&t=DvaAYjbkgvhHwCO0-1>